

ANR-JCJC **MIMICO**

Dr. Nicolas VERSTAEVEL

Exploring the synergies between
machine learning and agent-based modelling

02/2025 - 02/2028



MIMICO

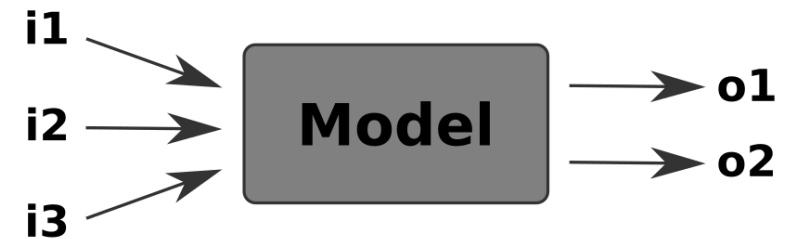
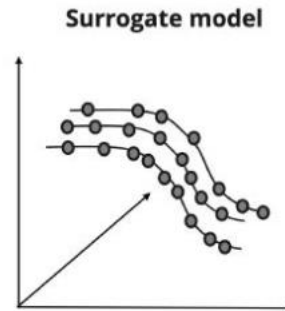
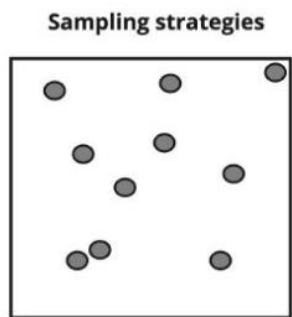
Problématiques et enjeux : validation

Question globale : Qu'est-ce que le MS nous apprend sur l'espace des paramètres ?



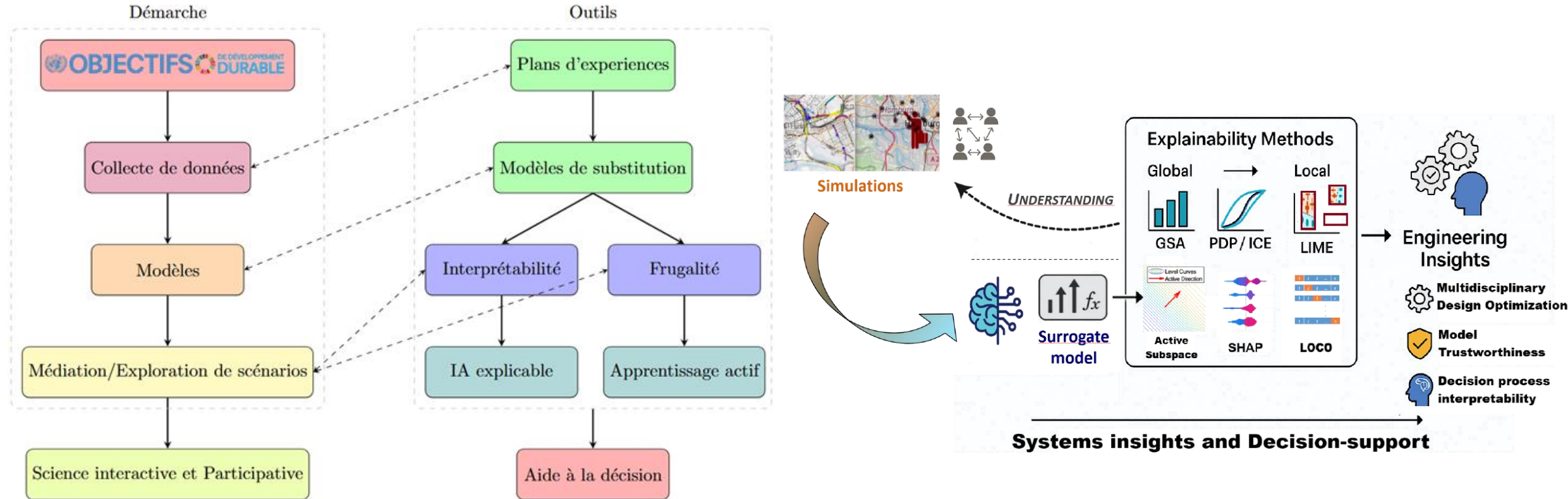
Q1 : Qu'est-ce que le MS nous apprend sur l'espace des paramètres ?

Q2 : Comment le MS nous informe-t-il sur la dynamique du modèle ?

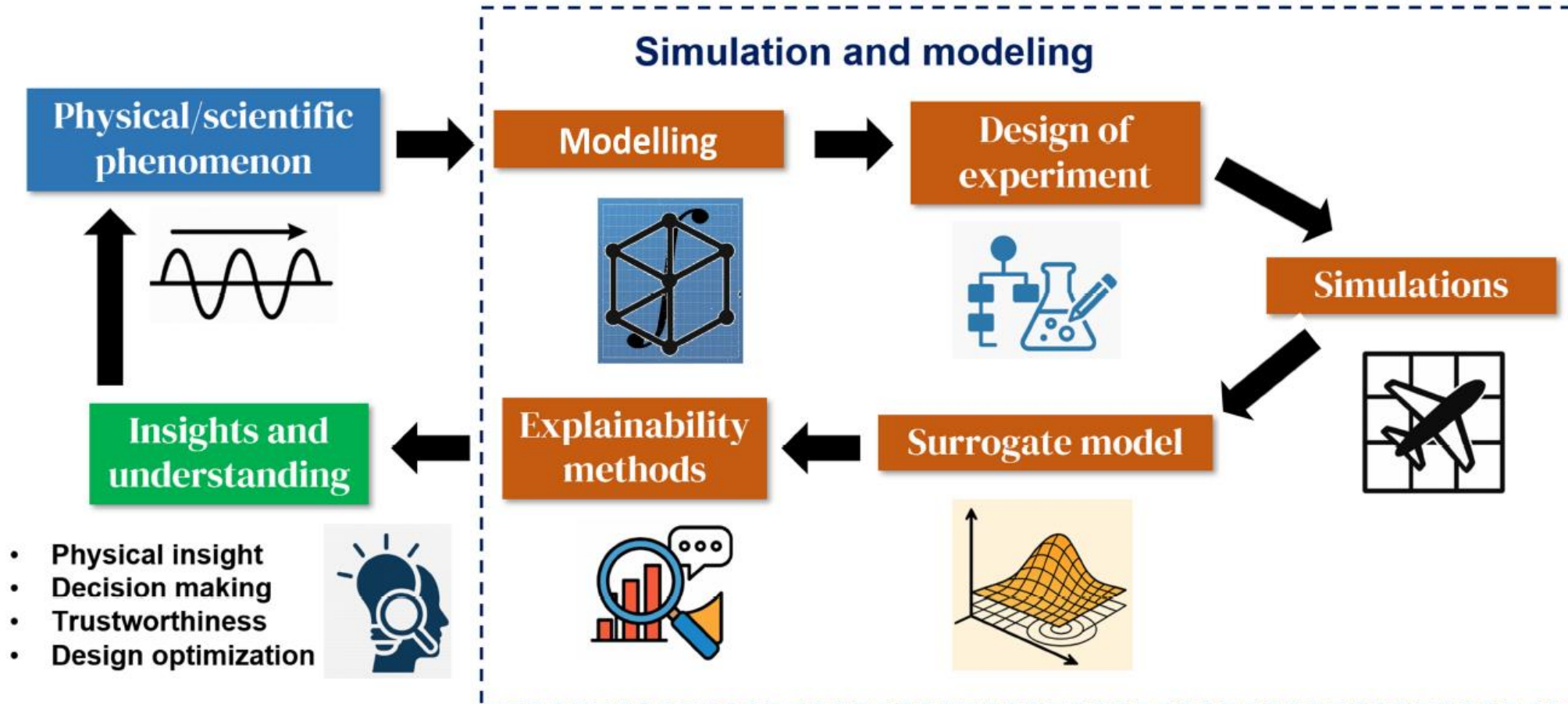


Q3 : Quel modèle pour quelle application ?

Modèles de substitutions pour l'aide à la décision



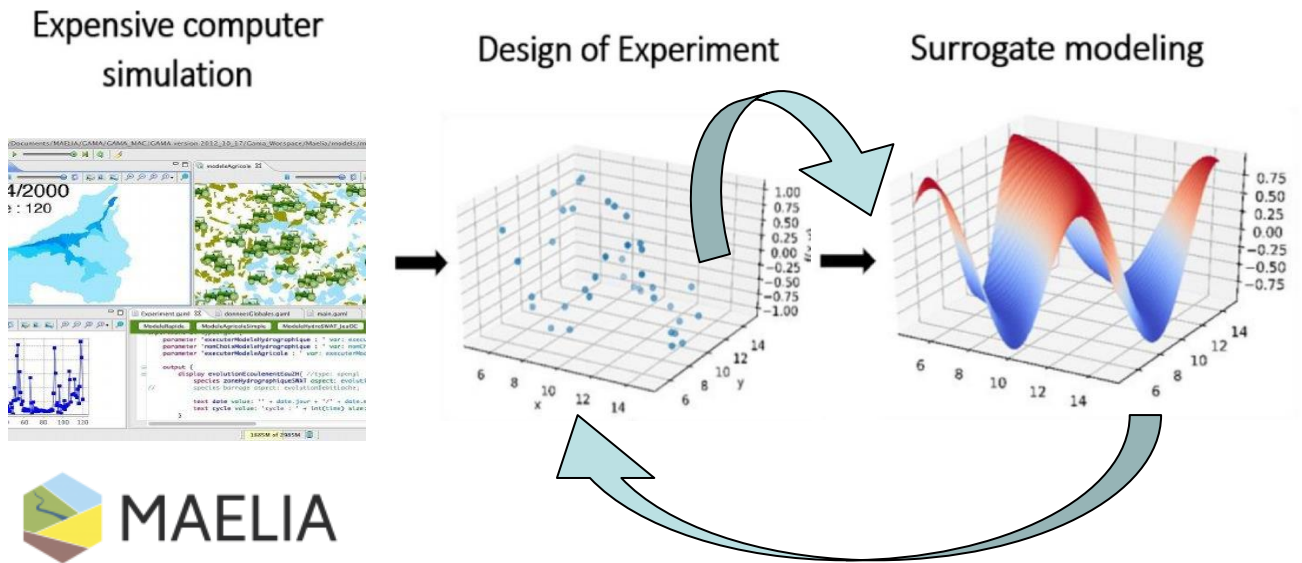
Méthodologie associée à nos approches



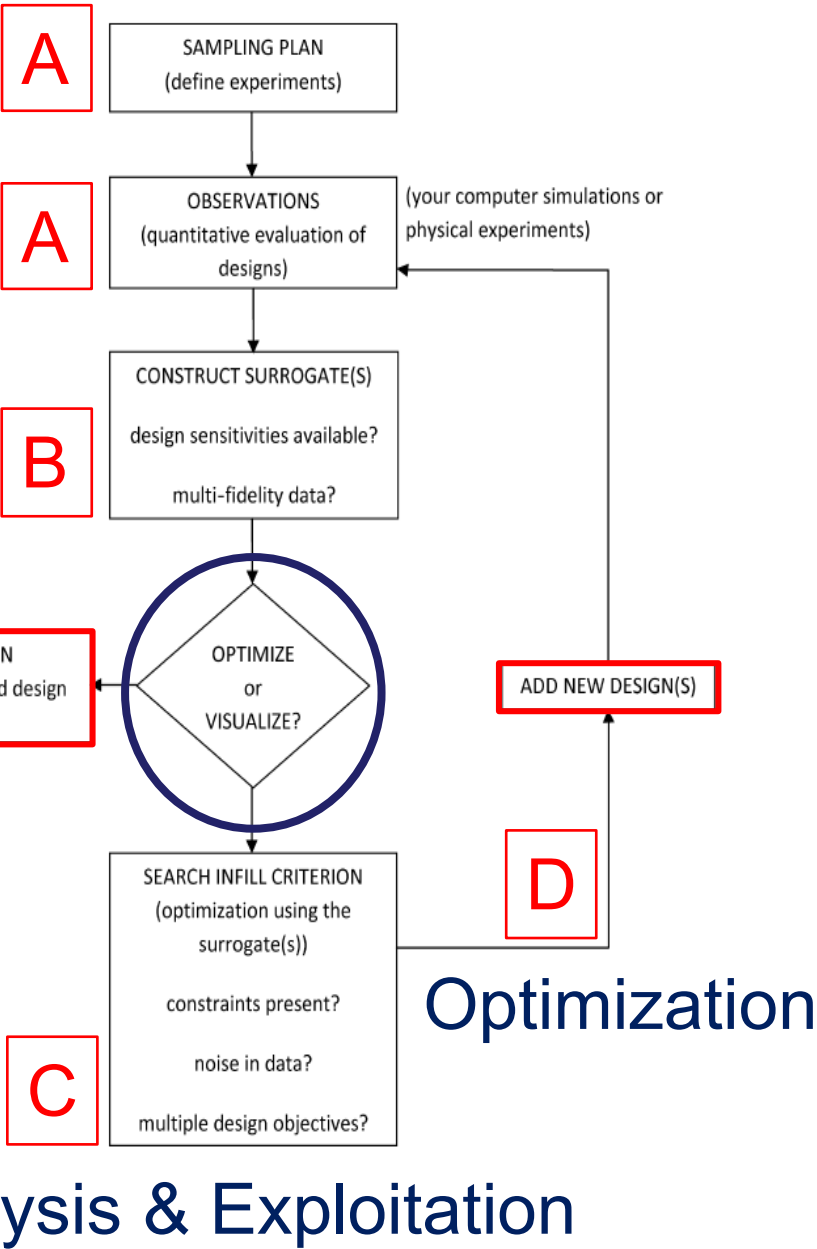
Surrogate modeling

Sampling

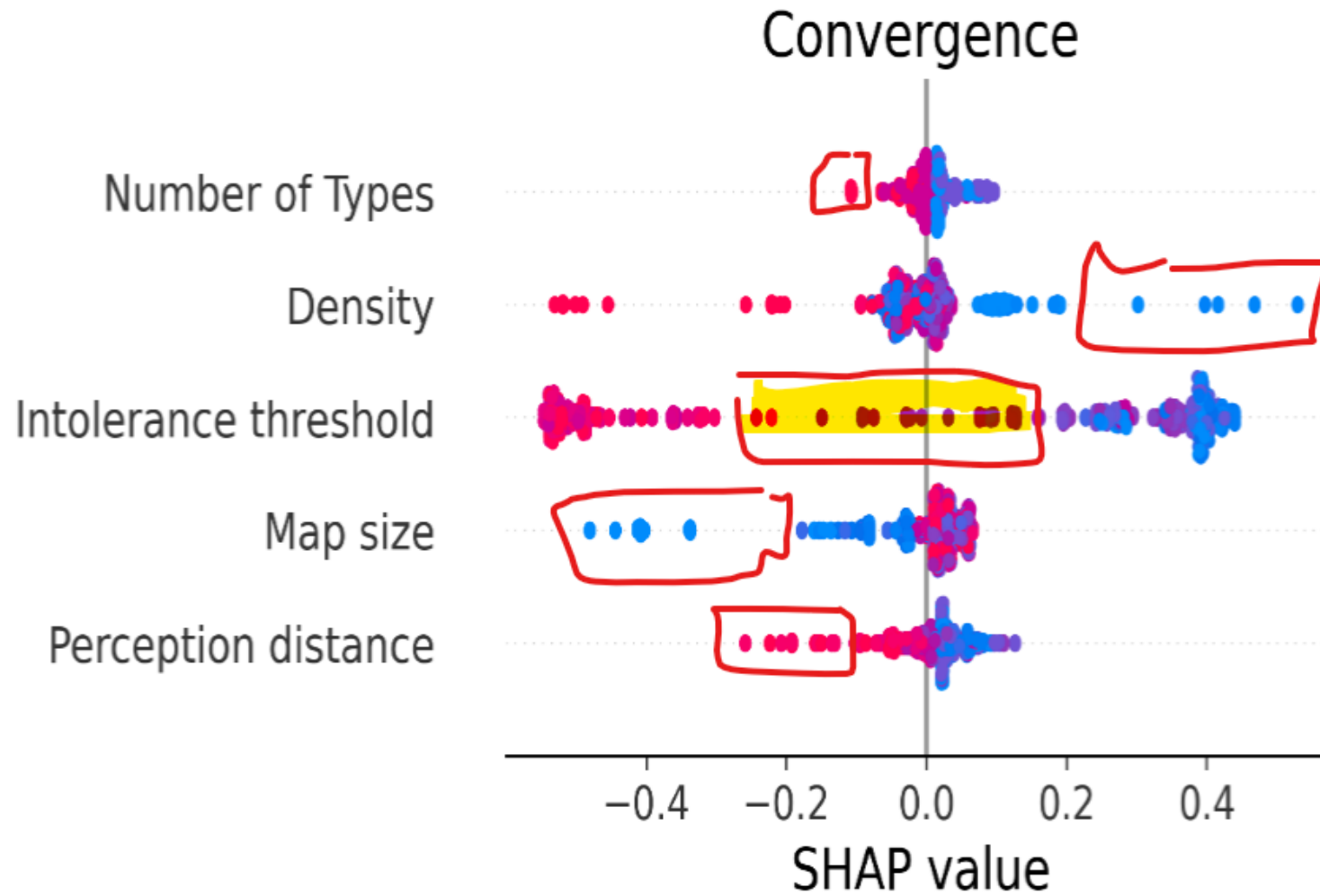
Models



Bayesian optimization: Adaptive sampling



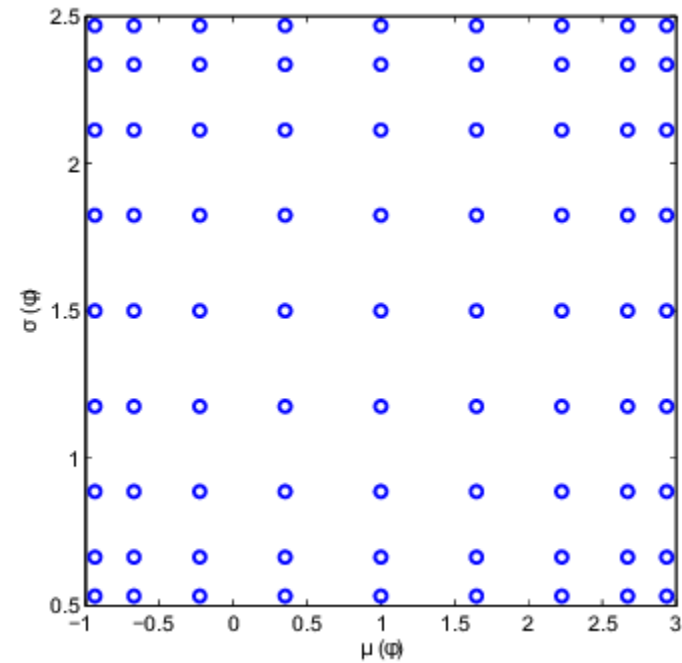
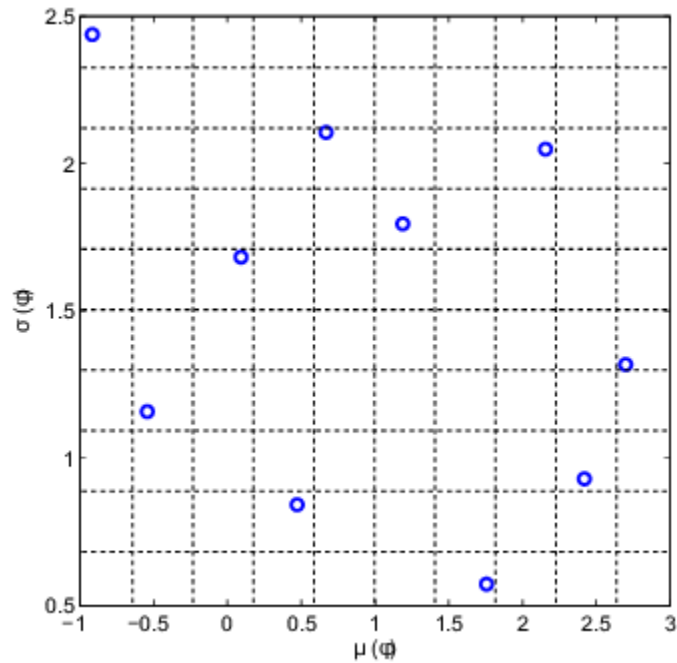
Motivation and main idea



1) Initial Dataset

Sample n_{doe} points DoE (LHS) in the features design space $:= x_{\text{doe}}$

Evaluate the expensive black box n_{doe} times $:= y_{\text{doe}}$



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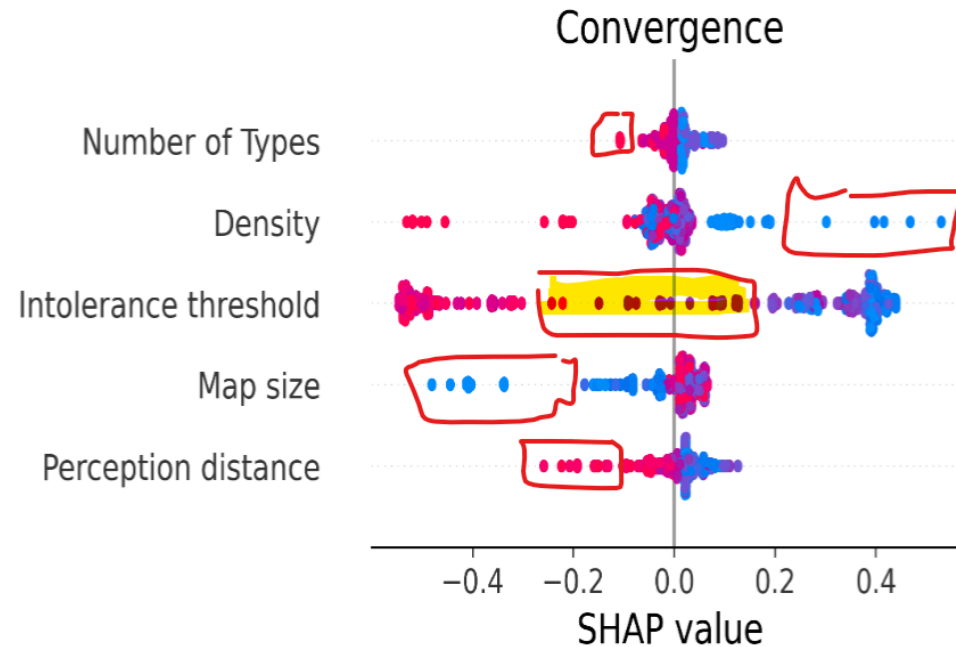
For i in $n_{\text{doe}} \dots n_{\text{point_total}}$:

2) Explanation inputs

Sample $n_{\text{xpl_lowfi}}$ points in the features design space $:= x_{\text{xpl_lowfi}}$

Create a noisy Gaussian process based on $\{x_{\text{doe}}, y_{\text{doe}}\}$

Compute ExactSHAP on $(x_{\text{xpl_lowfi}} \cup x_{\text{doe}})$ points. Reference given by x_{doe} . Using the GP



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3) Explanation densities

Compute density on $\text{SHAP}(x_{\text{xpl_lowfi}} \cup x_{\text{doe}})$ by KDE $:= y_{\text{dens_total}}$.

$y_{\text{dens_high}} := y_{\text{dens_total}}\{x_{\text{doe}}\} / y_{\text{dens_low}} := y_{\text{dens_total}}\{x_{\text{xpl_lowfi}}\}$

Convergence

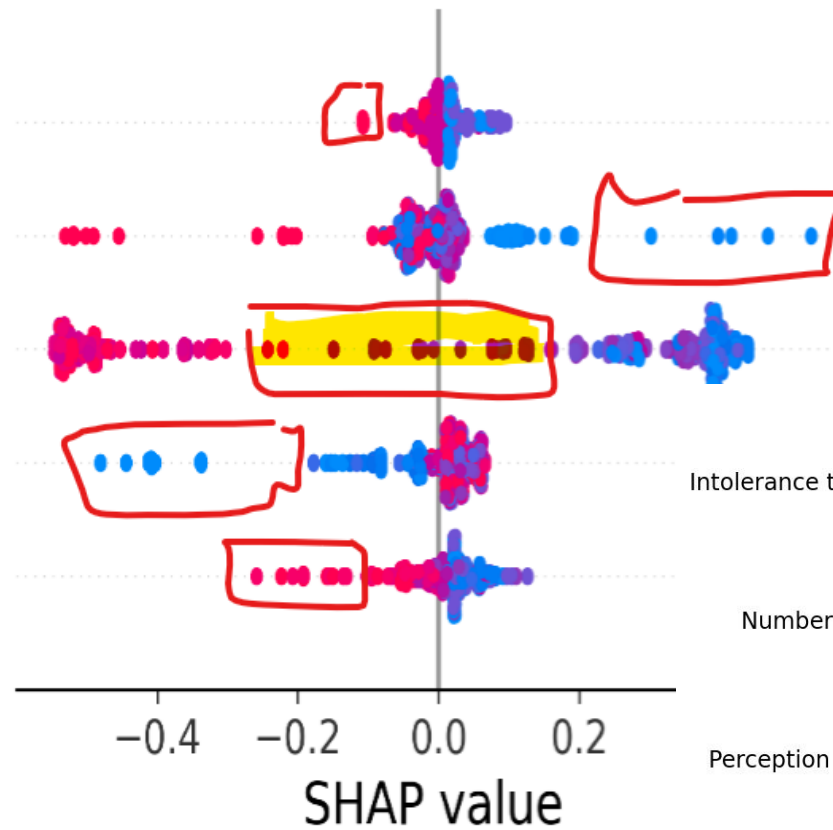
Number of Types

Density

Intolerance threshold

Map size

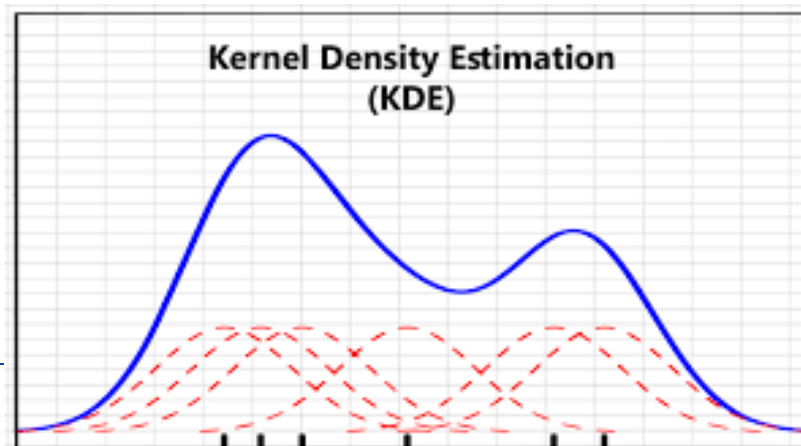
Perception distance



3) Explanation densities

Compute density on $\text{SHAP}(x_{\text{xpl_lowfi}} \cup x_{\text{doe}})$ by $\text{KDE} := y_{\text{dens_total}}$.

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SHAP Beeswarm Plot

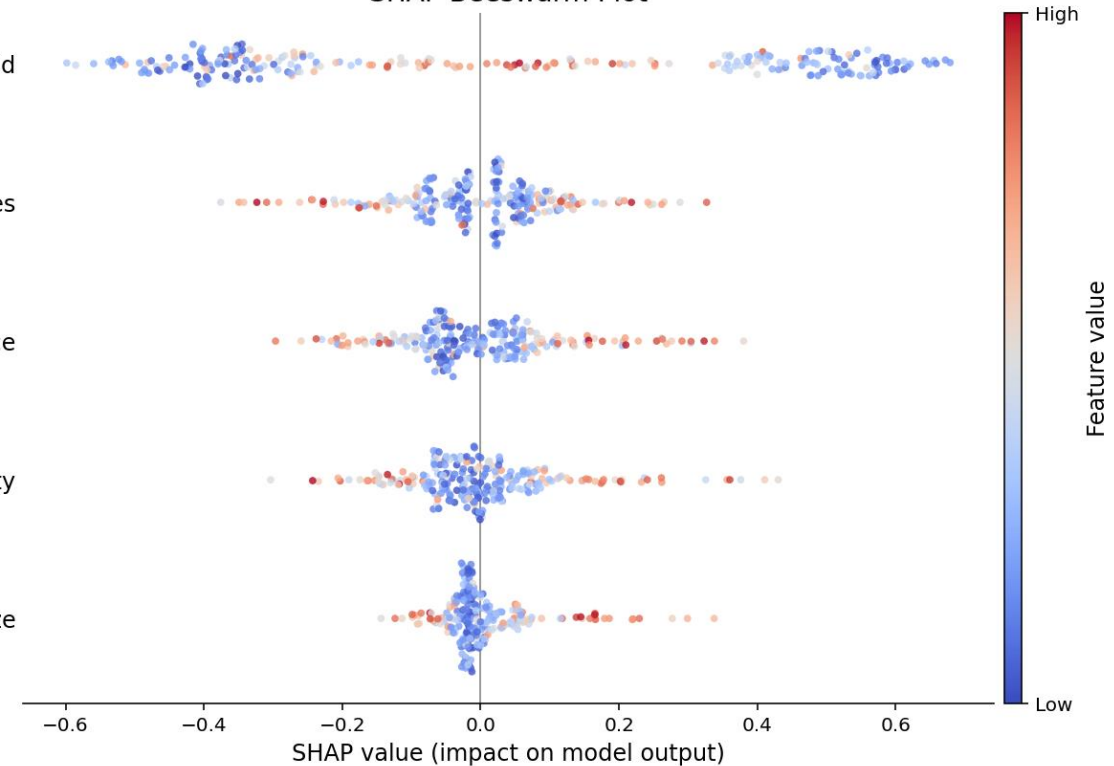
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4) Multi-fidelity Bayesian optimization

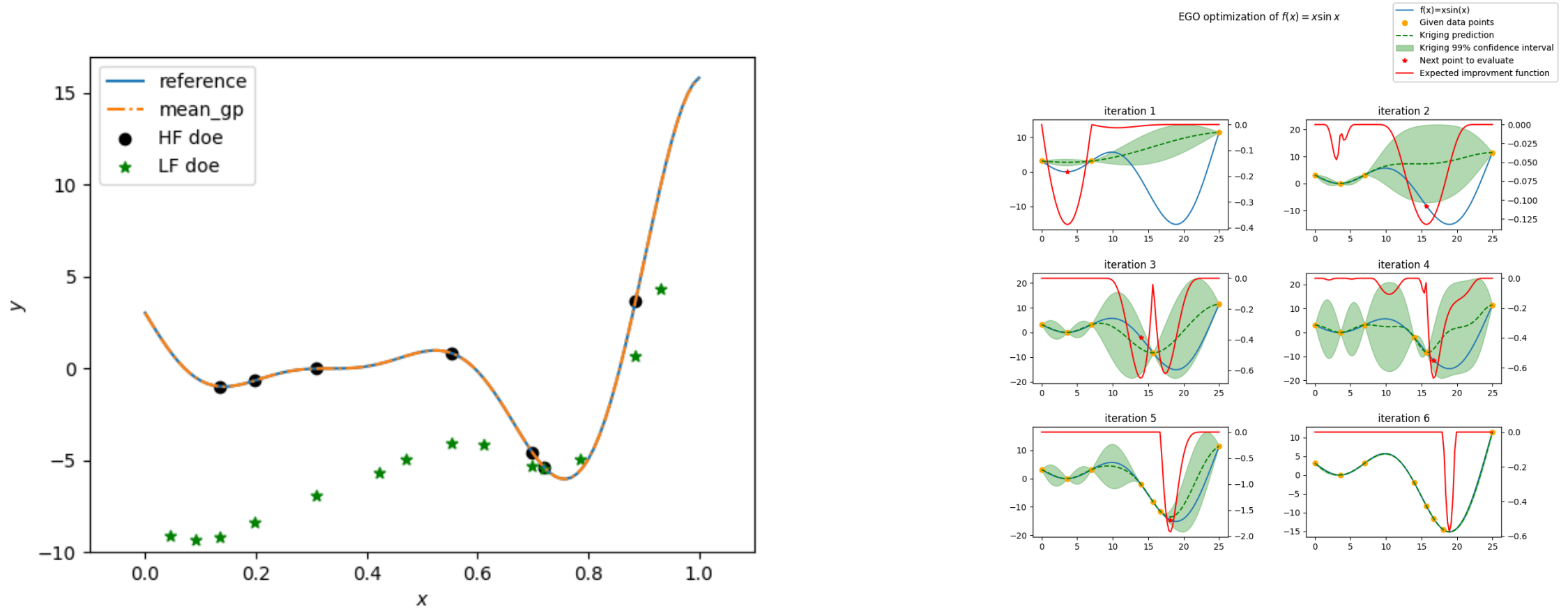
Create a noisy Multi-fidelity Gaussian process based on $[\{x_{\text{doe}}, y_{\text{dens_high}}\}, \{x_{\text{xpl_lowfi}}, y_{\text{dens_low}}\}]$

Optimize the Expected Improvement based on the MFGP $:= x_{\text{opt}}$

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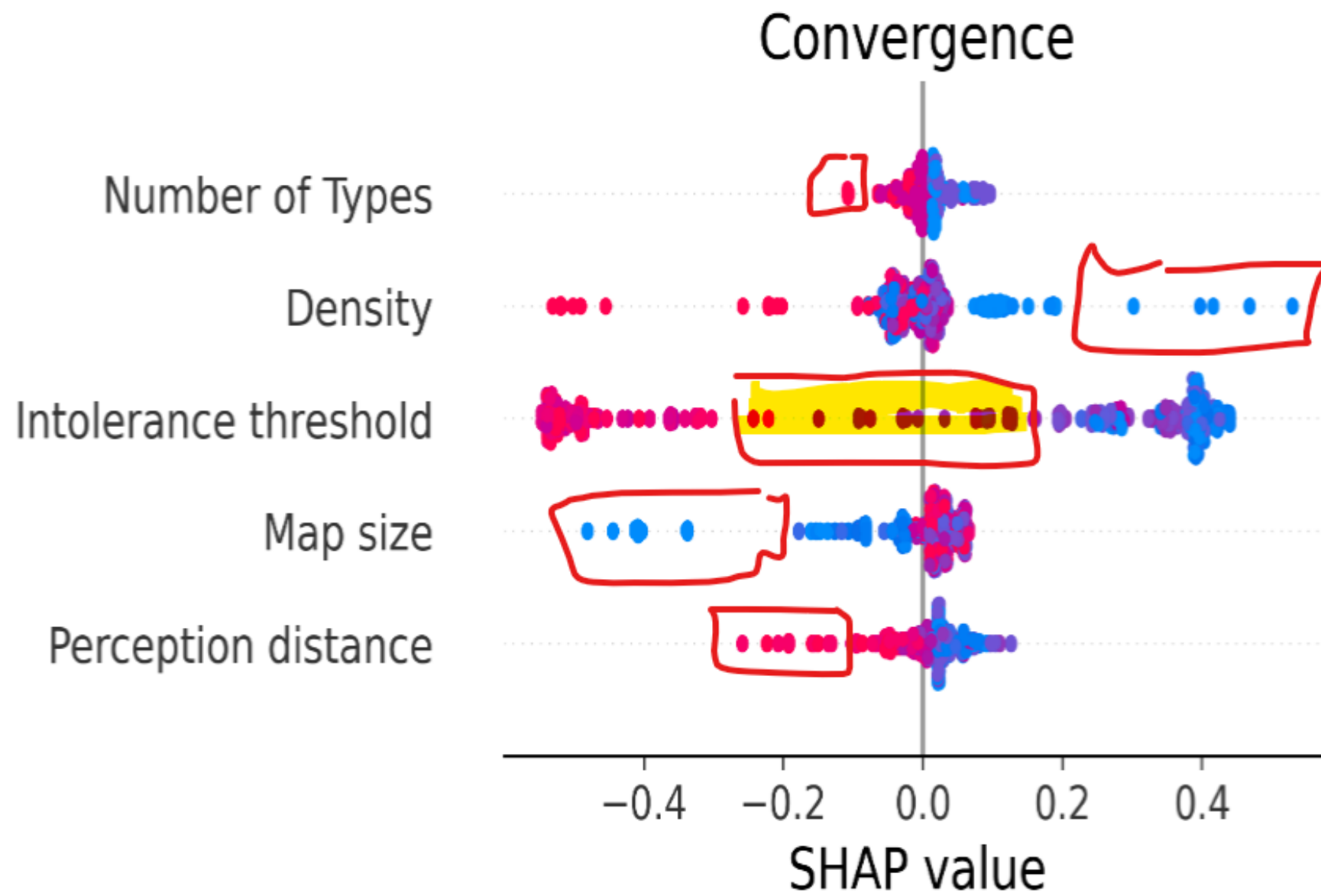
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5) Update

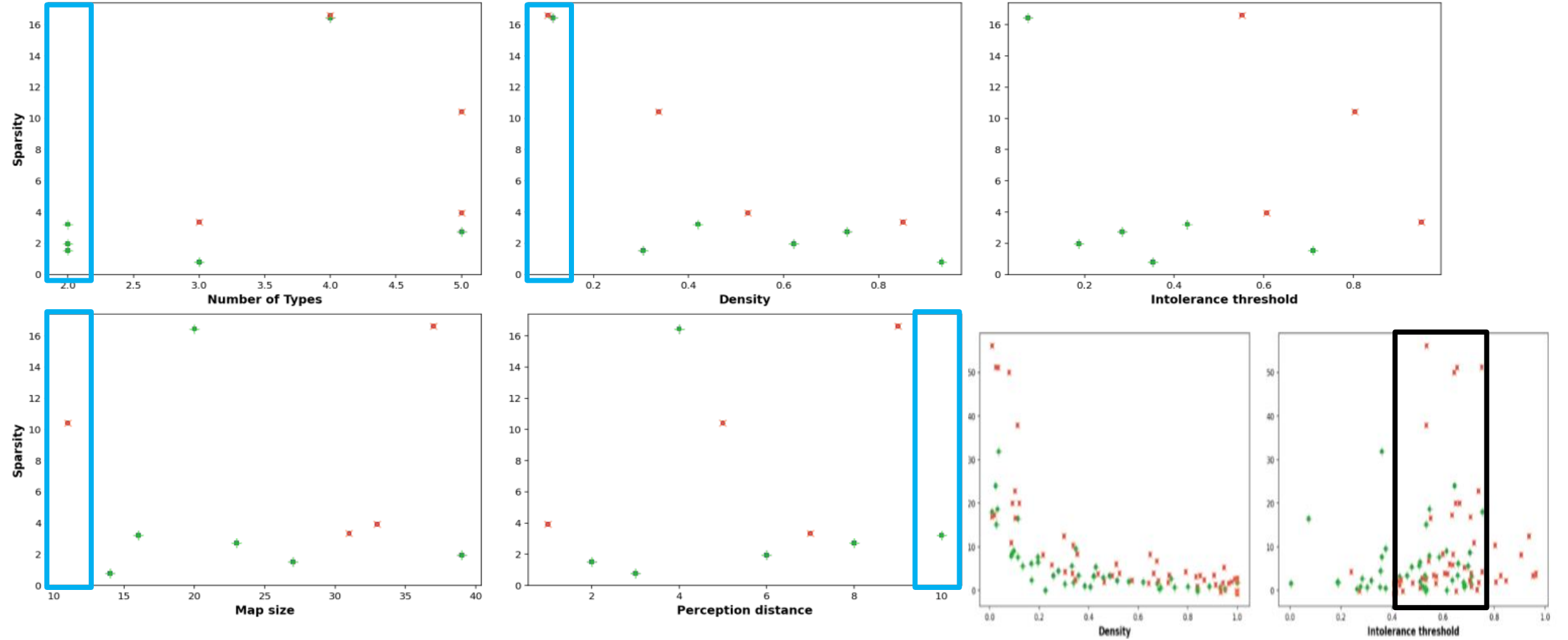
Evaluate the expensive black box in $x_{\text{opt}} := y_{\text{opt}}$

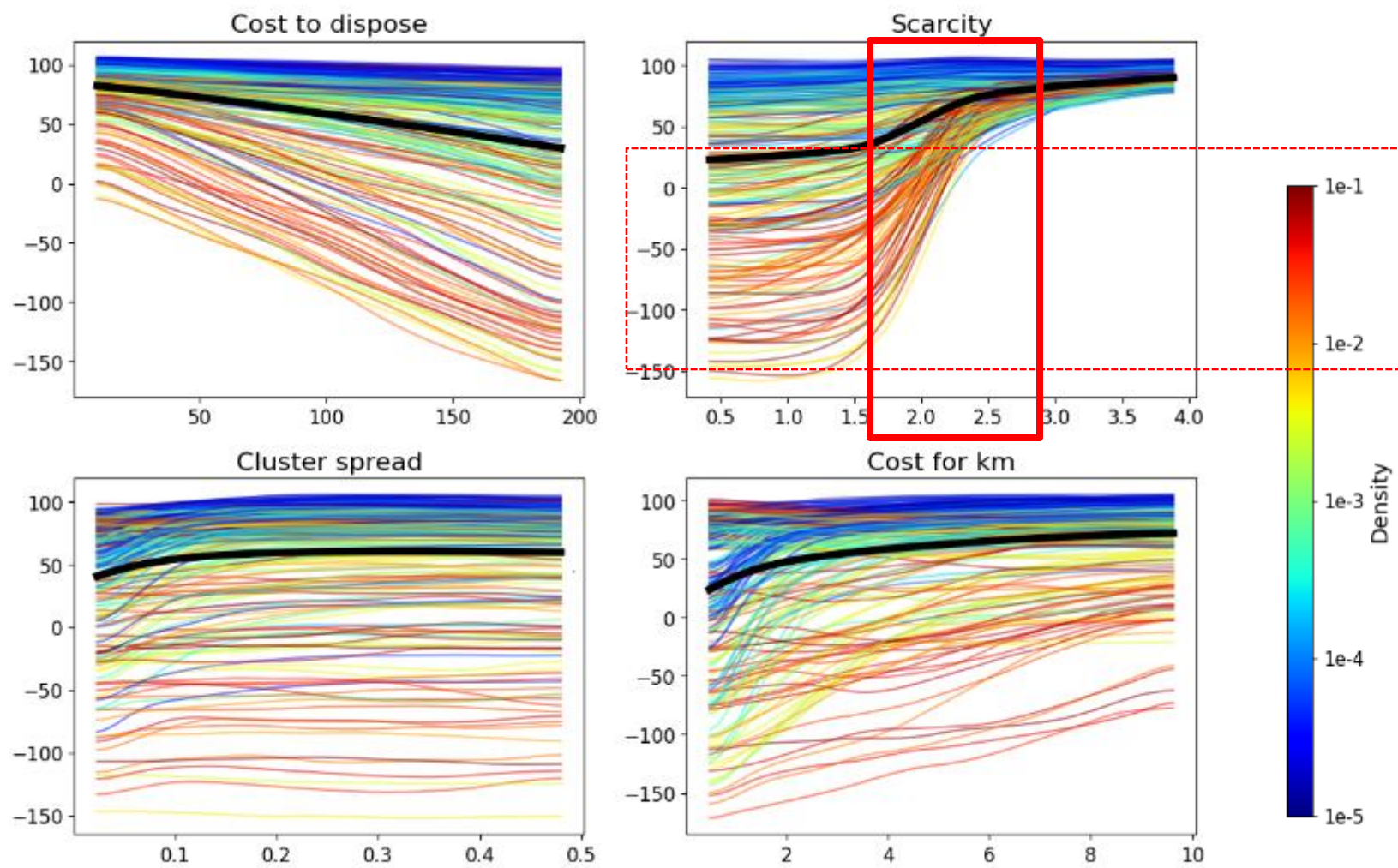
$x_{\text{doe}} := (x_{\text{doe}} \cup x_{\text{opt}})$

$y_{\text{doe}} := (y_{\text{doe}} \cup y_{\text{opt}})$



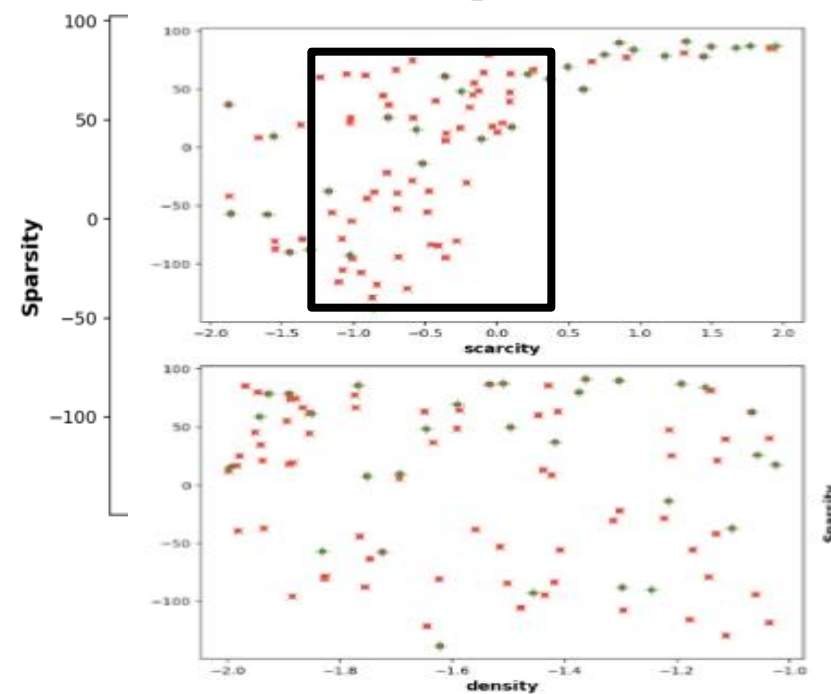
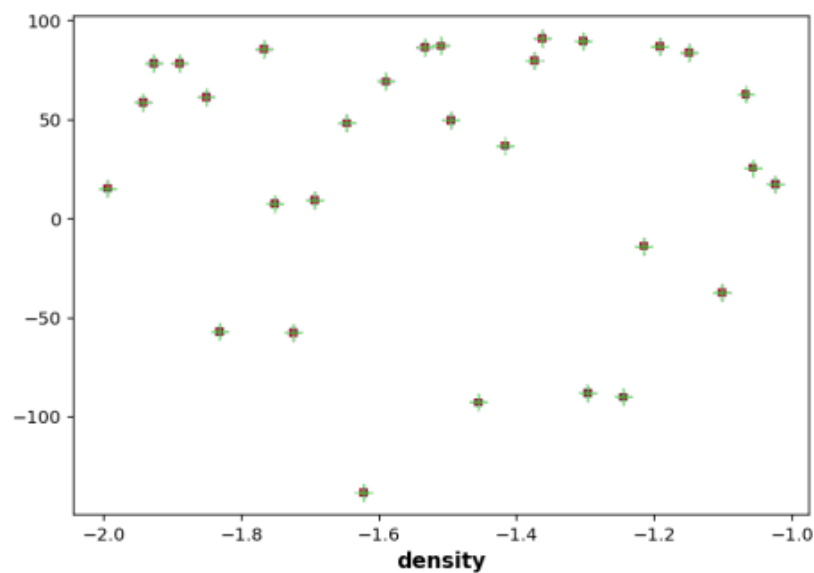
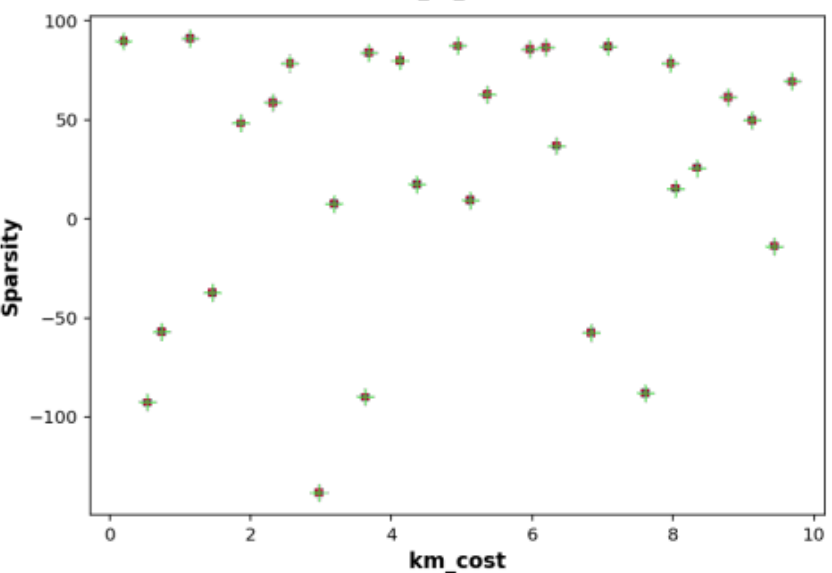
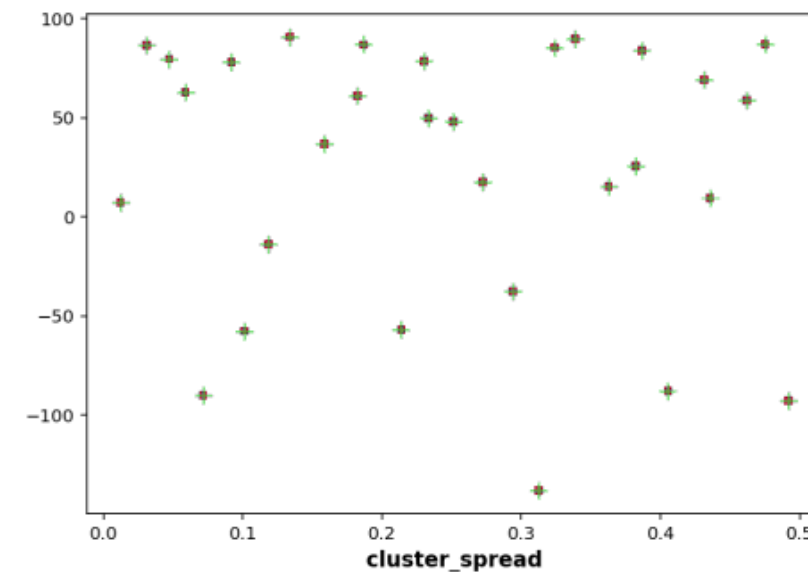
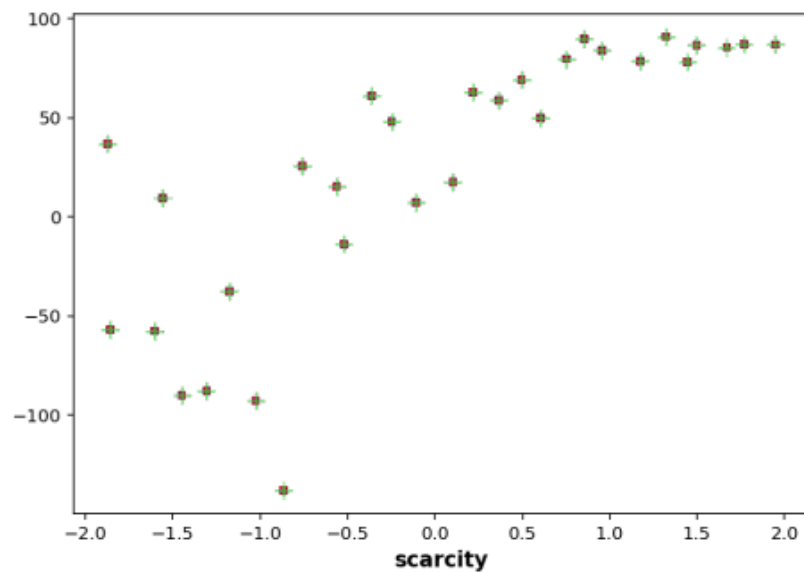
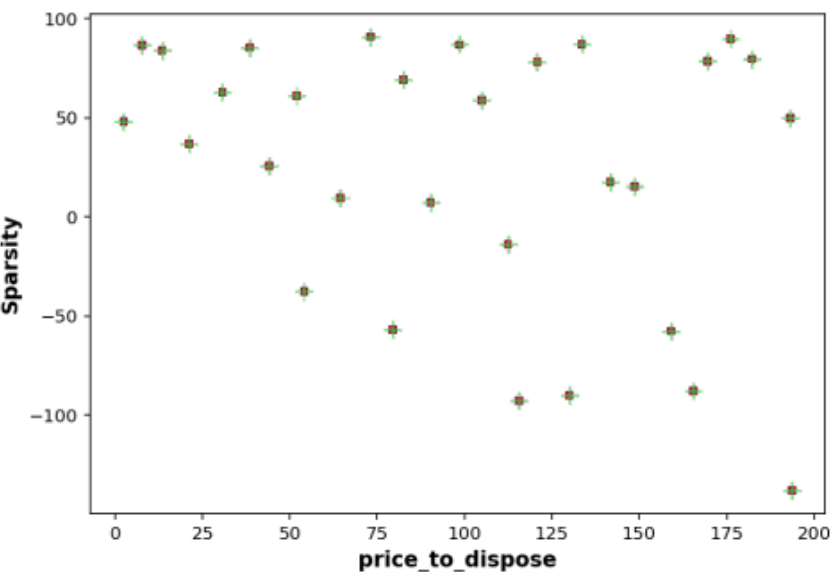
Evolution of Sparsity vs Features (N=10/100)

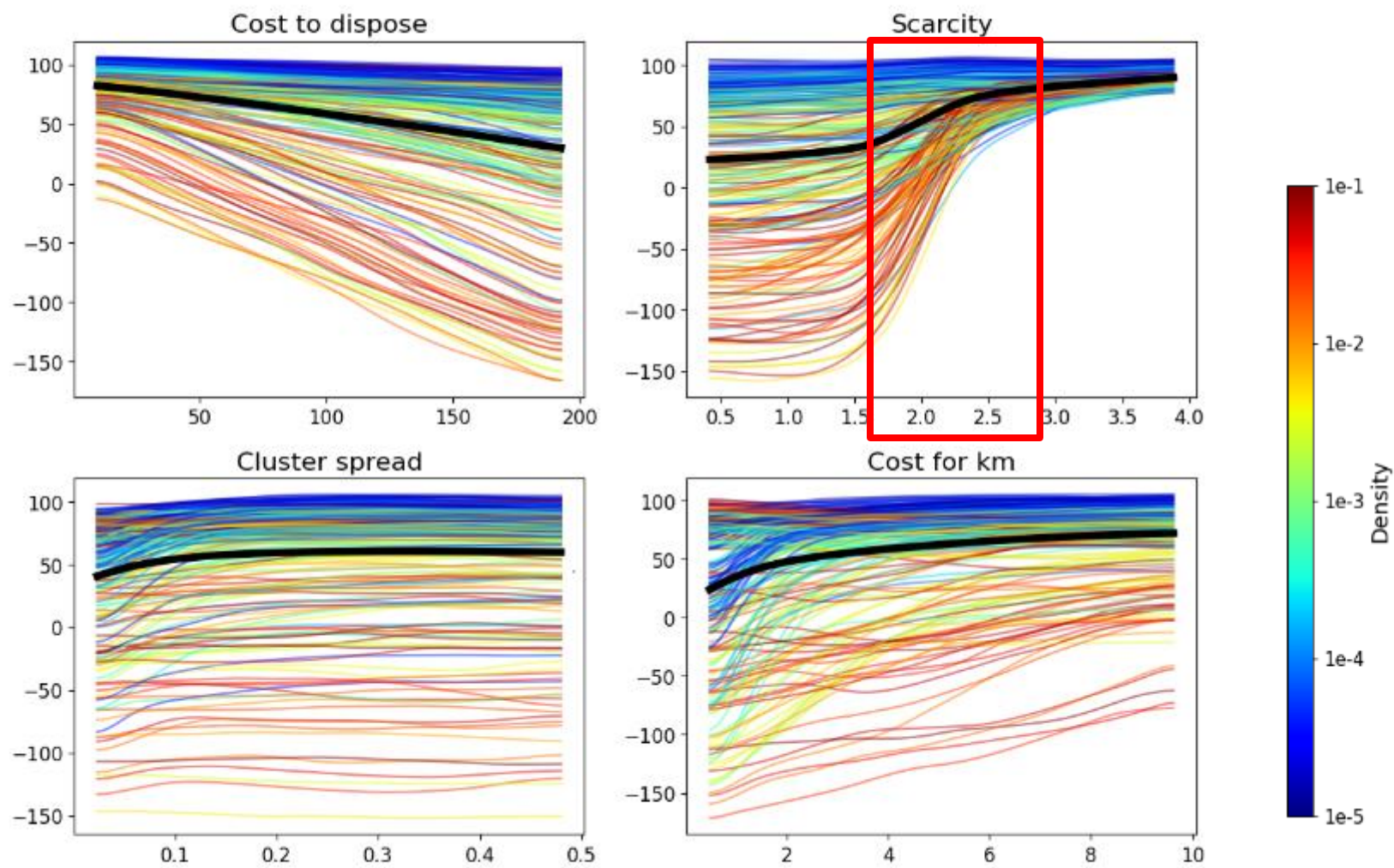




(a) PDP and ICE for the price index

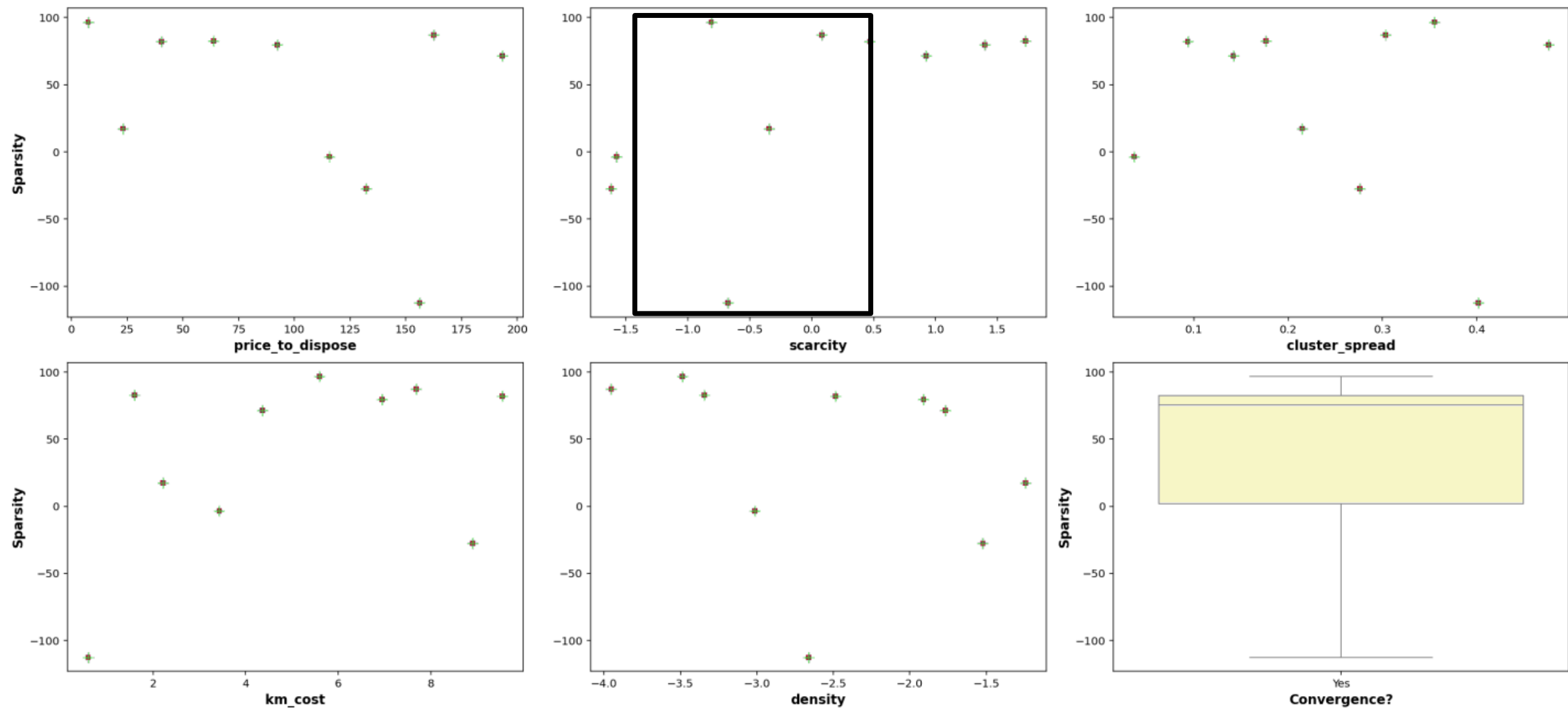
Evolution of Sparsity vs Features (N=30/100)





(a) PDP and ICE for the price index

Evolution of Sparsity vs Features (N=10/150)



Next steps

- Quantify the density shift during optimization => Where were the points added
- Add input replications // Improve the Bayesian optimization under noise
- Also, I worked with MLP instead of the simulator for testing => to update
- Add other test cases
- Write the paper
- Comparer avec d'autres algos => SHAP plus uniforme ! Montrer si on peut détecter les tipping points ? Mieux que d'autres algorithmiques
- Classifier performances, 2D map frontier?